

Suggested Teaching Guidelines for
IT Infrastructure and Management and DevOps
PGCP-ITISS February 2026

Duration: 50 Theory hrs + 40 Lab hrs + 30 Self-Learning hrs (Total: 120 hrs)

Objective: To introduce the student to Concepts of ITIL and Data center management.

Evaluation Method: Theory exam – 40% weightage
Lab exam - 40% weightage
Internal assessment – 20% weightage

List of Books / Other training material

Courseware: Cloud Computing Black Book, Kogent Learning, Kailash Jayaswal, (Wiley), 2024,

A. ITIM (10T + 3SL)

a. Introduction to ITIL

Session 1: (2T)

Theory:

Service Strategy

- General strategy
- Competition and market space
- Service provider types
- Service management as a strategic asset
- Key process activities,
- Financial management
- Service portfolio management
- Demand management
- Key roles and responsibilities of staff

Session 2: (2T)

Theory:

Service Design

- Design of architecture, processes, policies, documentation, and allowing for future business requirements
- Service Design Package (SDP)
- Service catalog management
- Service level management
- Designing for capacity management
- IT service continuity
- Information Security

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Session 3: (2T)

Theory:

Service Transition

- Service Asset and Configuration Management
- Transition Planning and Support
- Release and deployment management
- Change Management
- Knowledge Management
- key roles of staff

Service Operation

- Balancing conflicting goals (e.g. reliability Vs cost etc)
- Event management
- Incident management
- Problem management
- Event fulfillment
- Asset management
- Service desk
- Technical and application management
- Key roles and responsibilities for staff

Session 4: (2T + 2SL)

Theory:

Continual Service Improvement (CSI)

- Training and awareness
- Ongoing scheduling
- Roles created
- Ownership assigned

Self-Learning: (SL based on Session 1-4)

- Learn how service strategy concepts and how IT services create business value
- Read about different service provider types (internal, external, shared services) with enterprise examples.
- Understand service design principles covering architecture, processes, policies, and future scalability.
- Learn the purpose and structure of a Service Design Package (SDP)
- Understand capacity management, IT service continuity, and information security in service design
- Understand service transition processes including change, release, and configuration management
- Find out the importance of knowledge management in smooth service transitions
- Find the purpose of CSI in improving service quality and efficiency

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- Learn how feedback and metrics drive continuous improvement cycles

b. Data Center Management:

Session 5: (2T + 1SL)

Theory:

Introduction

- Data Center Architecture, Requirements & prerequisites
- Required Physical Area for Equipment and Unoccupied Space
- Required power to run all the devices
- Required cooling and HVAC
- Required Network bandwidth
- Selecting a Geographic Location
- Guidelines for Planning a Data Center

Infrastructure in a Data Center

- Modular Cabling Design
- Points of Distribution
- ISP Network Infrastructure and WAN Links
- Network Operations Center and Monitoring
- Data center physical security, Logical security, and cleaning

Self-Learning:

- Study data center design principles such as raised floor design, structural planning, and vandalism protection
- Understand budget constraints, capacity planning, and characteristics of a highly reliable data center design
- Learn data center consolidation concepts and why organizations consolidate infrastructure
- Study data center security: physical, logical, network, and source-level security guidelines
- Understand system administration best practices, automation, disaster recovery, and operational continuity

B. DevOps (40 T+ 40L + 27 SL)

Session 6: Basics of Virtualization (2T + 1 SL)

Theory:

- Introduction of virtualization
- Virtualization types: type1, type2
- Virtualization, Hardware Virtualization, Para-Virtualization, Cloning, Snapshot, and Template

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Self-Learning:

- Understand why virtualization exists and its role in DevOps and cloud platforms
- Compare Type-1 vs Type-2 hypervisors with real-world usage (data centers vs desktops)
- Learn snapshots, cloning, templates for rapid environment creation
- Understand hardware vs para-virtualization and performance implications

Session 7: Storage Area Network (2T + 1SL)

Theory:

- Configuring a SAN (FreeNAS)
- Using SAN for high availability
- ZFS Volume Configuration
- IP-Based Storage Communication

Self-Learning:

- Understand the role of shared storage in high availability and CI/CD systems
- Learn ZFS concepts (snapshots, integrity, pooling) used in enterprise storage
- Understand IP-based storage (iSCSI) and its relevance in private clouds
- Learn why storage design impacts deployment reliability and recovery

Session 8: Cloud Computing (2T + 3SL)

Theory:

- Introduction to Cloud
- Cloud Computing, Cloud SPI Model, Cloud Computing Deployment Model (Public, Private, and Hybrid), Cloud Security (SLA and IAM).
- Cloud architecture
- Service models: IaaS, PaaS, SaaS
- Services provided by Cloud (Compute, Database, Developer Tools, Storage, Media, Mobile, Web, Security, Integration, etc.)
- Cloud development best practices
- Learn how cloud API integration is done

Self-Learning: (3SL)

- Understand cloud computing fundamentals and why DevOps relies on cloud platforms
- Learn IaaS, PaaS, SaaS from an automation and CI/CD perspective
- Understand cloud architecture and commonly used services (compute, storage, security)
- Learn IAM, SLA, and cloud security basics required for DevOps

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engineers

Session 9 & 10: Infrastructure Provisioning using Chef and Puppet (4T+ 4L + 2SL)

Theory:

- Chef Basics
 - ◆ Chef Architecture
 - ◆ Chef installation
 - ◆ Chef Usage
- Puppet Basics
 - ◆ Puppet Architecture
 - ◆ Puppet Installation
 - ◆ Puppet Usage

Lab Assignment:

- Configuring Chef to perform infrastructure provisioning
- Configuring Puppet to perform infrastructure provisioning

Self-Learning: (2SL)

- Understand configuration management and why manual server setup does not scale
- Learn Chef and Puppet architectures and their role in infrastructure automation
- Understand desired state configuration and idempotency
- Gain awareness of legacy vs modern DevOps tooling trends

Session 11: Network Monitoring using Prometheus (2T + 4L + 2SL)

Theory:

- Prometheus and its concepts
- Prometheus installation
- Prometheus usage
- Prometheus monitoring

Lab Assignment:

- Install and configure Prometheus on a Linux server
- Monitor Linux systems using Node Exporter
- Monitor Windows systems using Windows Exporter
- Visualize system metrics using Prometheus Web UI

Self-Learning: (2SL)

- Understand monitoring as a core DevOps and SRE responsibility.
- Learn Prometheus architecture, metrics, and exporters.
- Understand system monitoring for CPU, memory, disk, and network.
- Learn how metrics support alerting, troubleshooting, and capacity planning

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Session 12, 13, 14 & 15: DevOps and Docker (6T + 4L+ 4SL)

Theory:

- Introduction to DevOps
- CI/CD Pipelines
- Docker concepts
- Docker commands
- Docker compose
- Docker Networking
- Volumes
- Docker Compose
- Tagging and pushing images to dockerhub

Lab Assignment:

- Create an httpd container running on port 8000. Name the container as web5. Do not map any directory. Create an “index.html” file & copy this file inside the container and check if the website displays your webpage. Again, modify the “index.html” file and copy it inside the container, and check if the website is updated or not.
- Create a C program(exe). Test the program on-base machine. Then create an image using Dockerfile to execute this exe program. Give image name as “Name: v1”
- Try the above with a program that requires user input

Self-Learning: (4SL)

- Understand DevOps principles, culture, and CI/CD pipeline stages.
- Learn Docker fundamentals: images, containers, volumes, and networking.
- Understand Docker Compose for multi-container applications.
- Learn image tagging, versioning, and publishing for deployment pipelines
- Learn about agile, lean
- Find out about the agile methodologies: Scrum and Kanban
- Find out how Lean and agile are used in Devops

Session 16 & 17: Git and Github (3T + 4L+ 1SL)

Theory & Lab:

- Introduction
 - ◆ Introduction to Git
 - ◆ Core Concepts
 - ◆ Going Command Line
- Basic Git workflow with Github
 - ◆ Welcome to GitHub
 - ◆ Setup the Project Folder
 - ◆ Git Configuration (User Name and Email)

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- ◆ Copy the Repository from GitHub to Your Local Computer (git clone)
- ◆ The First Commit
- ◆ Publishing Changes back to GitHub (push)

Self-Learning: (1SL)

- Understand Git as the foundation of DevOps workflows.
- Learn GitHub collaboration workflows used in teams
- Explore Github Actions vs Jenkins

Session 18 & 19: Jenkins (4T + 4L + 2SL)

Theory:

- Introduction of Jenkins
- Installing Jenkins
- Creating simple jobs
- CI/CD pipeline using Jenkins
- Integrating jenkins with github
- Integrating Jenkins with Docker

Lab Assignment:

- Create a C program to take input from the user and test it. Then create an image using Dockerfile. Test image by the running container. Then push the image to the docker hub repository. Then also export the image in a “cprog-back.tar” file.
- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the NGINX image.
- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the httpd image.

Self-Learning: (2 SL)

- Understand CI concepts and Jenkins role in automation
- Learn how code changes trigger pipelines
- Read about Jenkins architecture and pipeline concepts
- Learn build, test, and deploy automation using pipelines
- Learn how Jenkins integrates with Docker for deployments
- Create a Jenkins Pipeline: Github -> Jenkins -> Docker Build -> Run Container

Session 20 & 21: AWS (4T + 5L + 3SL)

Theory:

- Introduction to AWS
- Services provided by AWS: EC2, Lambda, S3
- Introduction Virtual Private Cloud (VPC) Setup

Lab Assignment:

- Create AWS EC2 instance

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- Create AWS Lambda
- Create AWS S3 bucket
- Create AWS VPC
- Create a new VPC by the name ditiss-lab with public & private subnet. Assign network address as 172.20.0.0/16. Assign 172.20.5.0/24 to the public and 172.20.10.0/24 to the private subnet. Create one instance and connect to a private subnet. Create one instance and connect to the public subnet. Install httpd on the private instance and check with curl from the public connected instance.

Self-Learning: (3SL)

- Understand AWS global infrastructure and DevOps usage model
- Learn core services: EC2, S3, Lambda for application deployment
- Read about VPC basics including public and private subnets
- Learn cloud networking concepts essential for secure deployments
- Launch EC2 -> Deploy webserver -> Store logs in S3
- Explain VPC isolation and security

Session 22 & 23: Container Orchestration (5T + 5L+ 3SL)

Theory:

- Container Orchestration: Kubernetes, Dockerswarm
- Micro Service Deployment

Lab Assignments:

- Create a new docker image using NGINX. Push it to docker hub. Create a service on Dockerswarm. Keep initial replicas as 1 and then increase the replicas to 10. Reduce the replicas to 2. Create a new image by modifying "index.html". Push to docker hub and then update the service for a new image.
- Do the above using Kubernetes
- Configure a Linux as a Master node using Kubernetes and make 2 to 3 worker node clusters.

Self-Learning: (3SL)

- Understand why container orchestration is mandatory in production
- Learn Kubernetes and Docker Swarm fundamentals
- Understand microservices deployment and scaling
- Learn concepts of replicas, rolling updates, and self-healing

Session 24 & 25: Ansible (3T + 5L + 2SL)

Theory:

- Introduction to Ansible and Configuration Management
- Setting Up Ansible Environment
- Ansible Playbooks and YAML Basics
- Managing Ansible Inventory
- Ansible Roles and Reusability

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Lab Assignments:

- Setting Up Ansible Environment
 - ◆ Install Ansible on a local machine or a virtual machine.
 - ◆ Configure SSH keys for passwordless authentication to a target node.
 - ◆ Verify the setup by running the `ansible -m ping all` command.
- Writing and Executing Ansible Playbooks
 - ◆ Create a simple Ansible playbook to install and start a web server (e.g., Apache or Nginx) on a target node.
 - ◆ Use the playbook to ensure the web server is running and serving a default webpage.
 - ◆ Verify the setup by accessing the webpage from a browser.
- Managing Ansible Inventory and Roles
 - ◆ Create a static inventory file listing multiple target nodes.
 - ◆ Develop a role to install and configure a database server (e.g., MySQL or PostgreSQL).
 - ◆ Apply the role using an Ansible playbook to deploy the database server on one of the target nodes listed in the inventory.

Self-Learning: (2SL)

- Understand agentless automation using Ansible
- Learn playbooks, YAML syntax, and inventory management
- Understand roles and reusable automation patterns
- Learn how Ansible supports configuration drift control

Session 26 & 27: Terraform (3T + 5L + 3SL)

Theory:

- Introduction to Infrastructure as Code (IaC) and Terraform
- Setting Up the Terraform Environment
- Writing and Organizing Terraform Configuration Files
- Terraform State Management
- Terraform Modules and Reusability

Lab Assignments:

- Setting Up the Terraform Environment
 - ◆ Install Terraform on a local machine.
 - ◆ Configure AWS credentials for Terraform to interact with AWS services.
 - ◆ Create and apply a basic Terraform configuration to provision an EC2 instance in AWS.
- Writing and Organizing Terraform Configuration Files
 - ◆ Write a Terraform configuration file to provision a VPC, subnet, and an EC2 instance within the subnet.
 - ◆ Use variables and outputs to make the configuration more dynamic and reusable.

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- ◆ Apply the configuration and verify the resources are created in AWS.

Lab Assignments:

- Terraform State Management and Modules
 - ◆ Create a Terraform module for provisioning an EC2 instance with a security group.
 - ◆ Use the module in a Terraform configuration to provision multiple EC2 instances in different subnets.
 - ◆ Manage the Terraform state using a remote backend (e.g., AWS S3) to enable collaboration and state sharing

Self-Learning:

- Understand Infrastructure as Code (IaC) and why it is critical in DevOps
- Learn Terraform workflow: plan, apply, destroy
- Understand state management and remote backends
- Learn how modules enable reusable and scalable infrastructure